



Society of Petroleum Engineers

SPE-229034-MS

Unlocking Faster-Well Delivery Opportunities Through a Smarter Engineering Approach, Utilizing Tandem Hole Opening for the First Time in the Region, a Case Study

Mohamed Osama, Gamal Salem, Ali Sulaiman Bin Sumaida, Abdulla Bin Shamlan, Saeed Almazrouei, Alauddin Abou Draz, Saleh Al Ameri, Haya AlSabi, Faisal Siddiqui, and Mouza Al Shehhi, ADNOC Onshore; Bassem S. El Yossef, ADNOC Upstream; Ahmed Maher Ayad, Nabors

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228859-MS](https://doi.org/10.2118/228859-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents a pioneering regional application of **Tandem Hole Opening (THO)** to accelerate well delivery in a geologically complex and operationally sensitive drilling environment. The challenge addressed was chronic borehole instability and extended rig time caused by conventional multi-run hole enlargement practices, particularly in salt-shale formations known for reactive behavior and poor hole cleaning. Historically, these formations required sequential drilling and reaming runs, increasing exposure time and operational risk.

Comprehensive pre-job engineering including torque-and-drag simulations, hydraulic analysis, and vibration modeling was carried out to design a BHA capable of progressive enlargement from a pilot hole to final diameter in one pass. During execution, real-time monitoring of key parameters such as weight on bit, rate of penetration (ROP), downhole shocks, and tool cutter wear was undertaken to ensure operational safety and efficiency.

To overcome these limitations, a **dual-hole opener Bottomhole Assembly (BHA)** was engineered to simultaneously drill and progressively enlarge the borehole from 8½" to 17½" in a single run eliminating the need for dedicated hole-opening passes. This approach was implemented for the first time in the region as part of a strategic underground oil storage project, where well delivery speed and borehole integrity were critical. Extensive pre-job engineering including torque and drag simulations, vibration modelling, and hydraulic analysis was conducted to validate the feasibility of the **THO BHA**. The design featured a staged enlargement profile using fixed-type hole openers positioned near the bit and higher up the string, enabling smoother cutting dynamics and improved hole cleaning. During execution, real-time monitoring of drilling parameters allowed adaptive control, ensuring safe and efficient operation.

The results were significant: the **THO** technique delivered a **47% reduction in section drilling time**, saving **36 rig hours** and enabling the well to be completed **several days ahead of schedule**. Additionally, the operation avoided **\$3.9 million in tool and service costs** compared to conventional underreaming

methods. Caliper logs confirmed a uniformly enlarged borehole, and casing was run to bottom without issues, validating the mechanical and hydraulic success of the design.

Beyond operational efficiency, the THO approach contributed to environmental benefits, including reduced fuel consumption and lower emissions estimated at 318 tonnes of CO₂ avoided. The trial also demonstrated enhanced safety by minimizing trips and simplifying the operational sequence.

This successful field trial marks the first deployment of THO using this technique in the region, as per author investigation showcasing its viability as a smarter engineering solution for faster, safer, and more cost-effective well construction. This successful field trial demonstrates how a smarter engineering approach to hole enlargement can unlock faster delivery, reduce operational risks, and offer valuable lessons for future wells in similar complex formations. The paper concludes with key findings and recommendations for scaling this technique, positioning tandem hole-opening as a viable method to improve drilling performance in challenging environments.

Introduction

Drilling efficiency and well delivery times are critical performance metrics in the oil and gas industry. Operators continuously seek innovative methods to drill wells faster without compromising safety or wellbore quality. One area of improvement is in hole enlargement practices, especially in challenging formations that demand multiple drilling runs or special hole conditioning. Conventionally, when a well plan calls for a borehole larger than the pilot bit size, the typical approach is to drill a pilot hole and then perform a dedicated reaming run (hole opening) to enlarge the diameter for next casing which normally performed in two or even three runs. This sequential process, while effective, adds considerable rig time and exposes the wellbore to instability during the extra trip. In geologically challenging zones such as salt formation with shale impurities that tend to swell or slough into the wellbore – prolonged exposure and additional trips can exacerbate borehole instability, leading to stuck pipe or hole collapse issues.

Recent advancements in drilling tools have enabled hole-enlargement-While-Drilling (HEWD), where an underreamer is included in the drilling BHA and activated while drilling the section. This saves time by combining two operations into one. However, standard HEWD configurations still have limitations. In many cases, the underreamer is placed some distance above the bit (often above the measurement-while-drilling tools for logging purposes), which means a portion of the hole (the "rathole" just above the bit's deepest penetration) remains un-enlarged at section total depth. In similar case operators must then either live with a short under-gauge rathole or make a separate clean-out run to eliminate it, incurring additional time. Moreover, enlarging a hole section with a single underreamer can introduce significant vibrations and torque fluctuations, especially if the reamer is taking a large cut or reaming a long interval of previously drilled hole. These dynamics were noted in prior studies of overseas where dual-reamer use, which showed that splitting the reaming work can reduce shock loads.

To overcome these challenges, a tandem hole-opening technique was tailored by the operator in one of its development projects. THO refers to using two underreamers in the same BHA typically one positioned near the bit and another higher up to incrementally enlarge the wellbore in stages while drilling. This concept, sometimes called a dual reamer/hole opening BHA Technique, that has been introduced in complex overseas operations to eliminate ratholes, dedicate run to open the hole to accommodate final casing string and enhance borehole stability. By activating a near-bit reamer, the final few meters of hole can be enlarged at the same time as drilling, thereby removing the need for a dedicated run to open the hole to final hole size, traditionally the practice is to open the hole from 8 ½" to 17 ½" section it has to be done in two dedicated runs (from 8 ½" to 12 ¼" and then from 12 ¼" to 17 ½") instead, by utilizing the THO technique this can be done in one run only.

In the regional context of this case study, the driver for adopting THO was to accelerate well delivery in a high stakes drilling campaign. The well in question targeted a problematic salt-shale section known

for instability and slow drilling rates. Historically, offset wells in this field experienced trouble when reaming the salt including stuck pipe incidents and repetitive wiper trips to maintain gauge causing delays in running casing. The Asset Owner's drilling team identified tandem hole-opening as a "*smarter engineering approach*" to this problem, hypothesizing that it could both mitigate these challenges significantly compress the drilling schedule. To the authors' knowledge, this was the first implementation of a dual reamer/hole opening drilling assembly in the country, if not in the region that making it a pioneering attempt to introduce state-of-the-art drilling technology in the region.

This paper details the planning, execution, and outcomes of the tandem hole-opening operation.

The objectives were set clear which are:

1. to prove that a dual-reamer BHA can enlarge the wellbore in one run safely and reliably,
2. to quantify the time savings and drilling performance improvements achieved, and
3. to capture lessons learned for future applications.

In the sections that follow, we discuss the engineering design of the BHA and drilling program, the real-time operational considerations, and the post-run analysis of performance. We also compare the results to conventional approaches and similar case histories from the literature to contextualize the benefits. Finally, the paper provides conclusions on the viability of THO for wider use, along with recommendations to refine this technique.

Drilling Environment and Challenges

The trial took place in an onshore development well drilled by the operator. The well plan called for a large-diameter hole section to be drilled through salt interval of interbedded anhydrite and shale sticks. This interval is not challenging in drilling but the during the hole enlargement as well. Cuttings from this formation are also stickier and less freely transported, raising the risk of pack-off if hole cleaning is suboptimal. In previous wells, the operator managed these risks by drilling a smaller pilot hole through the salt section and then enlarging it with conventional drilling BHA. The added challenge now is that the drilling wells are adjusted in pad with 10m slot to slot spacing as part of a unique design of the underground oil storage project. The drilling campaign started of the first well followed the historical practise, which presented many challenges to the operation not only in drilling but for the future cavern design too, this put the team in position to overcome these challenges in upcoming wells. Pointing that each additional trip exposed the open hole for several more hours, during which it impacts the total well and project delivery.

The net effect was lost time and occasional wellbore complications, undermining the efficiency gains of any fast drilling through the salt section.

Key challenges identified for the new approach were:

- **Borehole Stability:** Maintaining hole integrity while reaming and drilling concurrently, especially with an enlarged diameter in the sensitive and problematic formation. With a larger open-hole size, effective super saturated salt mud support and inhibition would be even more critical to prevent sloughing and washout.
- **BHA Design Constraints:** Incorporating two reamers into the BHA without compromising its directional capability or the integrity of other tools (especially measuring instruments). The lower (near-bit) reamer placement had to be below the measurement-While-Drilling (MWD) sensors, this mandated using advanced activation methods (e.g., on-demand hydraulic or electronic actuation), as it will be mentioning later, MWD could be eliminated from the BHA as part of BHA improvement.
- **Hydraulics and Cuttings Removal:** Drilling with dual reamers would generate a higher volume of cuttings per foot of hole drilled, as the annulus is being widened significantly. The drilling fluid

program had to ensure sufficient carrying capacity (high enough annular velocity and gel strength) to lift both the pilot bit cuttings and the reamer cuttings out of the well. Additionally, pump rates and pressures would increase when both reamers are cutting, so surface equipment and the bit/reamer nozzles needed to be configured to handle these parameters without excessive ECD (equivalent circulating density) that could induce losses.

- **Mechanical Loads and Vibrations:** A dual-reamer assembly would distribute cutting loads between two tools, but at the same time, having multiple cutting structures could introduce complex dynamic interactions. There was a concern about torsional oscillation (stick-slip) or lateral vibrations if the reamers started to interact adversely with the wellbore (for instance, if one reamer temporarily jammed or had uneven cutter engagement). Excess vibration can damage BHA components or cause premature tool failure. Thus, predicting and managing BHA dynamics was a critical part of planning.
- **Tool Reliability:** Both underreamers needed to function reliably opening and closing on command and staying intact while drilling a long interval. Failure of one reamer could leave an enlarged portion incomplete or create an obstruction. Particularly, the lowermost reamer, being a novel placement in this context, had to be proven to withstand the drilling conditions (weight, rotation, and fluid flow near the bit) without malfunctioning.

By identifying these challenges early, the engineering team put in place specific measures (discussed below) to mitigate them. The goal was to ensure that the tandem hole-opening run would be executed smoothly and bring tangible performance improvements over the conventional method which is significantly achieved.

Engineering Design and Methodology

Planning for the tandem hole-opening operation involved multidisciplinary collaboration between the operator's drilling engineers, the directional drilling service provider, and the underreamer tool vendor. Past case studies of dual-reamer BHAs from the industry were reviewed to glean best practices with limited information. Notably, a service company used dual-underreamer system (used in deepwater Gulf of Mexico) and similar setups in the North Sea provided confidence on the approach that a properly engineered BHA could drill and enlarge simultaneously. Building on these insights, the team custom-designed a dual-reamer BHA tailored to the specific well requirements.

The approach is to examine and follow the same overseas experience by deploying the solution of THO by utilizing the underreamer, despite the contract absence. The operator approached deal to deploy the underreamers with management approval however, the author reached out different approach as the application is not full suit the underreamer application and high associated cost, instead we have explored the market looking for hole openers which are rarely found in smaller size as (17 1/2" and 12 1/4"), by deep search and follow up we could secure those sizes from local supplier.

Tandem Hole Opening BHA Configuration

The BHA objective is to open from 8 1/2" to 17 1/2" section and BHA consisted of a pilot bit (8 1/2 in. diameter) followed by a near-bit opener and an upper opener. The near-bit reamer was positioned just above the bit on the lower BHA sub, capable of expanding to approximately 12 1/4 in. diameter. About 50 ft further up the drill string (above the MWD and rotary steerable tool) the second reamer was placed, designed to expand to 17 1/2 in. the full gauge for the section. The beauty of this design of tandem reamer arrangement would not only create a borehole profile of 17 1/2 in. throughout, while drilling ahead with an initial pilot cut of 8 1/2 in. The staged enlargement (8 1/2" → 12 1/4" → 17 1/2") was chosen for two reasons: (1) to split the cutting work between two tools so that each underreamer removes a smaller annulus of formation (reducing strain and

torque spikes), and (2) to allow the upper reamer a smoother action since it would be reaming a hole that was already partly enlarged by the lower reamer, rather than starting from the small pilot diameter. Figure 1 & 2 illustrates the tandem hole-opening concept and BHA schematic (simplified for clarity).

Tandem Hole Opening



Figure 1—illustrates the tandem hole-opening concept and BHA schematic

Item #	Description	OD (in)	ID (in)	Gauge (in)	Weight (ppf)	Top Connection	Length (ft)	Total (ft)
1	8 1/8" Bull Nose	6.010	2.000	8.125	168.11	P 6-5/8" REG	2.52	2.52
2	Bit Sub	8.000	2.000		147.00	B 6-5/8" REG	3.00	5.52
3	X-Over	8.000	2.750		147.00	B 4-1/2" IF	4.00	9.52
4	7" Drill Collar	7.000	2.750		102.22	B 4-1/2" IF	30.00	39.52
5	X-Over	6.500	2.750		147.00	B HL150	3.00	42.52
6	P-P X-Over	8.000	2.750		147.00	P 6-5/8" REG	4.00	46.52
7	B-B X-Over	8.000	2.000		147.00	B 6-5/8" REG	3.00	49.52
8	P-P X-Over	8.125	2.750		147.00	P 6-5/8" REG	1.30	50.82
9	8 1/2" x 12 1/4" Hole Opener	8.250	2.810	12.25	114.35	P 6-5/8" REG	5.10	55.92
10	B-B X Over	8.000	2.000		147.00	B 6-5/8" REG	2.55	58.47
11	8 1/4" Drill Collar	8.250	2.810		171.05	B 6-5/8" REG	30.00	88.47
12	P-P X-Over	8.000	2.000		147.00	P 7-5/8" REG	1.30	89.77
13	12 1/4" x 17 1/2" Hole Opener	12.000	3.000	17.50	215.66	P 7-5/8" REG	6.33	96.10
14	X-Over	8.250	3.000		147.00	B 6-5/8" REG	3.57	99.67
15	8 1/4" Drill Collar	8.250	2.000		171.05	B 6-5/8" REG	30.00	129.67

Figure 2—illustrates the tandem hole-opening BHA

Both hole opener tools are opening (fixed type) not hydraulic as underreamer types and here the beauty of this design. The huge cost savings over the project period that estimated of \$3.9M compared to other underreaming technologies. The theory design of hole opener is that has the capability to cut through the formation, 40-60% of fluid diversion through the tool to allow better hole cleaning and for that reason the MWD were eliminated in future stage on the project as part of BHA optimization, Prior to running in the hole, the reamers were tested on surface to verify their functionality through routine surface testing practise.

Modelling and Simulations

A thorough simulation exercise preceded the field run:

- *Torque and Drag*: Using a wellbore friction model, the team simulated the expected rotating torque with one and both reamers cutting. For the dual-reamer case, calculations predicted a slight increase in steady-state torque when both reamers were engaged which is justified and understandable, but still within the rig's capability and below the over-torque threshold for the drill pipe connections. The friction factors were adjusted for an enlarged hole and verified to ensure that casing running would remain feasible after drilling (with reduced drag due to the smoother wellbore).
- *Vibrations*: A finite element drill string dynamics simulator was employed to analyse potential vibration modes of the BHA. The simulation indicated that placing stabilizers appropriately and using modern reamers with **anti-vibration features** (such as specialized cutter geometry and shock absorbers) would keep vibrations in check. The critical rotary speeds to avoid (resonant frequencies) were identified – the analysis showed that very high RPM could induce a coupled oscillation, so the plan was to use moderate rotary speeds and high flow instead to achieve ROP.
- *Hydraulics*: The drilling fluid was a salt saturated mud (+/-10-10.5 ppg), Na chloride system formulated to control salt washout and shale-strikes swelling. Hydraulic simulations were run to determine pump pressure and ECD with two reamers open. At 700-900 gpm pump rate. This was acceptable given the rig pump capacity and did not fracture the formation as per leak-off test data. Annular velocity in the enlarged hole was checked. Additionally, the team designed sweeps (high-viscosity pills) to be pumped at intervals to aid in cleaning the larger hole.

Operational Procedure

The execution plan for the THO was carefully scripted:

1. **Run-in-Hole with tandem hole opening**: The BHA was run to bottom with both hole openers (the 17½" tools passed easily through the 18⅝" casing). After tagging the bottom of the previously drilled 8 1/2½" hole (at the last casing shoe depth), circulation was established. Then, start smoothly not aggressively by a programmed sequence drilling parameter.
2. **Drilling with Dual hole opener**: With the bit and reamers ready, drilling commenced at a controlled rate. The top-drive rotary speed was set initially around 60-710 rpm with 10-20 weight on bit, based on the modelling guidelines. As the bit drilled ahead in new formation, the near-bit reamer engaged almost immediately to start enlarging to ~12¼", and the upper reamer cut the hole to full gauge soon after. The depth lag between the bit and upper reamer meant that the upper reamer was cutting rock that had been drilled and partially enlarged just moments before by the lower tools. This **progressive reaming** sequence ensured that no reamer was suddenly encountering uncut rock at full gauge, which helped avoid shock loading. ROP averaged 30-50 ft/hr on average, as expected, but quite satisfactory given both reamers were cutting concurrently. Mud logs showed a continuous return of salt cuttings, which appeared fine and plate-like, typical as pilot hole drilled initially to acquire coring data.
3. **Monitoring and Adjustments**: Throughout the operation, real-time downhole tool data (from the rotary steerable system and vibration sensors near the bit) were monitored at surface. Downhole vibration remained in a low-to-moderate range. Notably, when entering a particularly hard streak in the shale, a mild increase in torsional oscillation was observed – the mitigation was to slightly reduce weight on bit and increase rotary speed by ~5-10 rpm, which smoothed out the torque curve. This responsive adjustment was facilitated by having engineering specialists on call at the Remote Operations Center, who were watching the data feed. Additionally, standpipe pressure and pump rate were modulated occasionally to ensure cuttings were clearing; a few short flow pauses were done to

perform "pumps-off" settling checks and then high-flow flushes. No significant pack-offs or pressure surges were seen. The dual-reamer BHA drilled the entire shale interval in one continuous run.

4. **Completion of Section:** Upon reaching the section total depth (TD) as planned, the procedure was to **ensure the hole was fully at gauge** and then safely withdraw the tools short trip was made back to the last casing shoe to condition the hole during this wiper trip the wellbore exhibited smooth torque and drag trends, indicating a **uniformly enlarged wellbore**. There was no tight spot or ledge felt, an early sign that the tandem reaming achieved a good gauge hole. The BHA was then pulled out of hole to surface.

During the process, team communication was paramount. The drilling crew, mud engineers, and directional driller all aligned their actions with a pre-agreed **operations matrix**. This included contingencies (for example, criteria to stop and stabilize if torque exceeded a threshold, or if losses occurred). In practice, none of these contingencies were needed – the operation progressed without any major abnormality.

Results and Observations

The tandem hole-opening operation was completed successfully, marking a first-of-its-kind achievement in the region. The following were the key results and observations recorded:

- **Time Savings:**

The dual-reamer drilling saved a full round-trip and reaming operation. Compared to an offset well where the 17½" section was drilled in two stages (8 ½" to 12 ¼" and then 12 ¼" to 17 ½"), the new approach reduced total time in that section by approximately **47%**. In absolute terms, about **36 hours of rig time** were saved on this one interval. This contributed to the well being delivered several days ahead of the original plan. The elimination of the dedicated clean-out trip not only saved time but also removed the associated flat time of rigging up and down another BHA.

- **Hole Condition and Quality:**

Upon pulling out, a comprehensive suite of logs and gauge runs indicated an in-gauge, smooth wellbore. A calliper log run by wireline showed the 17½" section to be nearly uniform in diameter with no major washouts or ledges. The calliper readings closely matched the planned diameter, verifying that both reamers had done their job throughout the interval. The **improved hole quality** had immediate positive impact: the 13⅜" casing was run to bottom without issues, and cementing operations encountered low drag and low risk of channelling. In previous wells, extra reaming often led to slightly irregular hole enlargements; by contrast, the tandem reamer approach produced a high-quality borehole despite the problematic formation.

- **Drilling Performance:**

The average ROP in the section was 30-50 ft/hr, which is respectable for the given formation and only marginally lower than the offset pilot bit that had no reamer. This indicates that the dual hole opener did not dramatically hinder the penetration rate. In fact, any slight loss in ROP was offset by the continuous progress (versus stopping to ream later). There were no severe torque spikes; the real-time data showed only minor fluctuations corresponding to transitions between rock layers. Similarly, drag while pulling out the closed tools was low pick-up and slack-off weights were very consistent, confirming that no tight spots were left.

- **Hole Cleaning Efficiency:**

One notable observation was that **cuttings cleaning appeared more effective** than anticipated. The continuous enlargement ensured that cuttings were not accumulating in one place: larger cuttings created by the bit were immediately broken down further by the reamers or lifted by the high annular flow in the enlarged hole. The mud returns at surface were uniformly filled with medium-to-small cuttings, and no episodes of high torque or pump pressure (which might hint

at pack-off) occurred. The team attributed this partly to the progressive reaming by breaking the rock in stages, the cuttings were smaller on average and easier to carry out. Moreover, the decision to use high flow rates and periodic viscous sweeps paid off; no significant fill was found at the bottom of the hole on wiper trips. The successful removal of cuttings also plays into the improved borehole condition noted above.

- **Tool Performance and Wear Observed:**

Both underreamers functioned without any mechanical failures throughout the run. After pulling out, the tools were inspected. The near-bit reamer's cutters showed normal wear patterns consistent with drilling the abrasive shale, but no broken cutters or abnormal damage. The upper reamer was likewise in good shape. These inspections demonstrate the robustness of the dual-reamer BHA design. Additionally, vibrations remained low (downhole shock data recorded only mild vibration levels), which helped avoid fatigue damage. The absence of tool issues is a crucial validation, as it means the concept can be repeated without concerns of compromising expensive BHA components.

- **Safety and Risk Reduction:**

By needing fewer trips, the operation inherently reduced certain risks: less exposure of personnel to heavy lifting or pressure-testing operations for a second BHA, and less cumulative time where something could go wrong. From a well control perspective, having one continuous operation meant fewer stop-start events where swab or surge pressures could induce kicks or losses. Indeed, no kicks or loss events occurred. Also, the on-bottom drilling time was shorter, meaning the shale had less time to hydrate and destabilize likely a factor in why no wellbore collapse happened. Overall, the tandem approach not only saved time but also enhanced operational safety by simplifying the sequence of events.

- **Environmental and Emission Reduction Benefits:** Another positive outcome of the tandem hole-opening operation was a reduction in the environmental footprint of drilling activities. By completing the section faster and in a single run, the rig's engines operated for fewer hours, directly cutting fuel consumption and emissions. In this trial, total 15 business days of rig time were saved; assuming a typical land rig emits on the order of 318 tonnes of CO₂ avoided and 0.6 metric tonnes of Nox. Beyond carbon dioxide, fewer rig days also mean proportional reductions in other air pollutants (e.g., NO_x, SO₂) associated with diesel combustion. Moreover, the elimination of an extra clean-out trip reduced the total volume of drilling fluids and cuttings handled, minimizing emission. This streamlined approach aligns with the operator's sustainability goals it not only saved rig time and cost but also lowered the well's overall carbon footprint. The trial thus demonstrates not only drilling efficiency gains but also how innovative practices can contribute to emissions reduction in line with industry decarbonization efforts.

These results strongly indicate that the trial was a technical and operational success, meeting the primary objectives set out. The ability to drill and enlarge the hole in one run delivered a step-change improvement in efficiency for this section. Furthermore, the risk mitigations put in place (careful design, monitoring) were effective, as evidenced by the lack of incidents., [Figure 3](#) Show the comparison of the tandem hole-opening concept from well to another throughout the project scope.

It is worth noting that during this operation, the drilling crew and engineering team did face a learning curve, since this was the first such run. For example, managing the simultaneous reaming required close coordination in adjusting drilling parameters. Once the team observed how the wellbore responded, they fine-tuned the weight on bit and rotary speed to optimize ROP while keeping torque steady. This real-time optimization was crucial and was facilitated by having the right data at hand. The success of the operation hence underscores not just the tool technology, but also the importance of real-time decision-making and teamwork. The author and project senior drilling engineer of the project was on top of the full operation

throughout the critical parts of the run, communicating with the remote support team a practice that the operator found valuable and plans to continue for any such innovative operations, table:01 highlighting the operational, economic, and safety advantages of the tandem approach. The comparison underscores how integrating reaming into the drilling run significantly reduces rig time, improves borehole quality, and enhances overall efficiency setting a new benchmark for well delivery in complex formations.

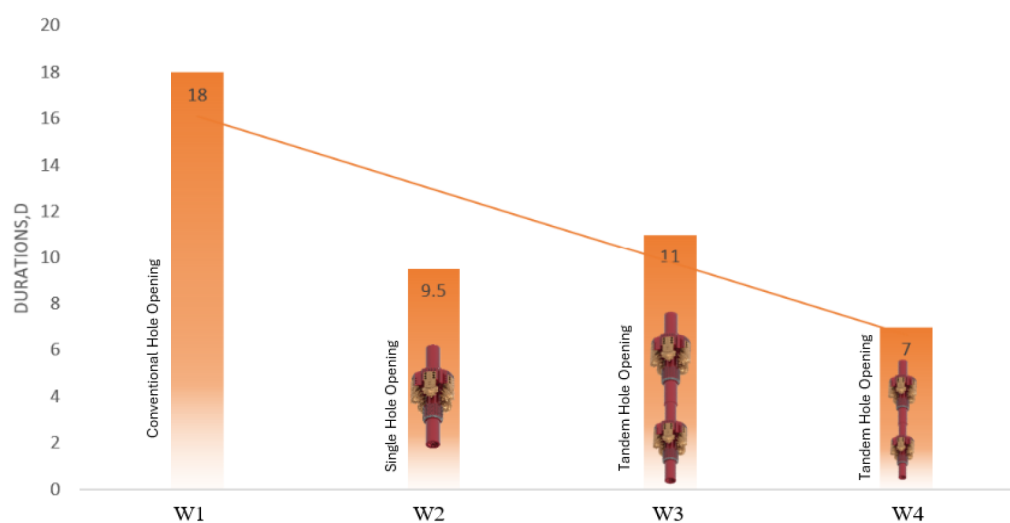


Figure 3—Show the comparison of the tandem hole-opening concept from well to another throughout the project scope.

Conventional vs Tandem Hole Opening

Table 1—Show constructive comparison of Conventional vs Tandem Hole Opening.

Metric	Conventional Method	Tandem Hole Opening	Improvement
Efficiency	Lower	Higher	47% reduction
Trips Required	2 (at least)	1	50% fewer
Cost	Higher (due to underreamer runs)	Lower (hole opener tools)	~\$3.9M saved
Hole Quality	Variable	Uniform	Improved
Operational Risk	Higher: more trips, longer exposure to unstable formations	Lower: reduced exposure, fewer stop-start events	THO minimizes borehole instability and stuck pipe risk
Tool Cost	High: underreamers used in multiple runs (~\$1M per well) as per current author operator practice	Low: fixed-type hole openers (~\$3.9M saved across project)	THO avoids premium underreamer service costs and use hole opener instead
ROP (Rate of Penetration)	Slightly higher during pilot drilling, but offset by reaming downtime	Slightly lower per foot, but continuous progress	THO achieves better net drilling efficiency
Vibration & Torque	Higher: single reamer reaming long intervals causes shock loads	Lower: distributed cutting load between two reamers	THO reduces mechanical stress on BHA
Environmental Impact	Higher emissions	Lower emissions	318 tonnes CO ₂ avoided
Safety	More exposure to lifting, pressure testing, swabbing	Safer: fewer trips, smoother operation	THO simplifies operational sequence
Learning Curve	Established practice	New technique; requires training and coordination	THO success depended on planning and real-time monitoring

Discussion

The outcomes of this case study demonstrate that THO can be a game-changer for drilling in challenging formations. Several discussion points emerge from the experience:

1. **Mechanism of Improvement:** The dual hole opening BHA achieved its intended effect by altering the mechanics of the drilling process. By breaking the removal of rock into two stages. The smoother torque profile not only protected the tools but also helped maintain a more constant ROP. Additionally, the MWD removed due to failure to operate the tool under the manufacture split flow that reached 40-60% inside the tool. This mechanism proved that it is functionality and way to reduce the associated drilling risks.
2. **Comparison to Conventional Approach:** Before this trial, the operator's conventional approach would involve drilling an 8½" pilot hole through the salt formation with shaly strikes (with minimal conditioning), then tripping out the bit and running a 17½" dedicated hole opener assembly to enlarge the hole using conventional drilling BHA. That approach faced difficulties on the first well to maintain between wells slot spacing (10m) as per design which led to change the hole opening approach on the rest project wells. In contrast, the tandem BHA effectively **merged these two steps**, ensuring the hole was enlarged before any significant time elapsed. The elimination of the intermediate time gap appears to be crucial in reactive formations. A direct outcome was that the casing could be run much sooner a strategy well-known to preserve hole condition. Thus, from a well construction viewpoint, the tandem method aligns with the best practice of "**drill and case quickly**", but accomplishes it without the extra reaming trip.
3. **Operational Complexity:** Introducing a second reamer and advanced activation systems did add complexity to the BHA and procedure. There were more failure modes to consider (e.g., what if one reamer failed?). In this trial, contingencies were in place. Fortunately, those contingencies were not needed. This raises a point: while tandem reaming provides clear benefits, it necessitates high confidence in tool reliability and operator proficiency. The drilling crew underwent special training sessions on the operation of the specific underreamers and the signs of proper functioning (pressure responses, etc.). The success here reflects not just the tools, but the preparation. We infer that for other operators considering this technique, a thorough risk assessment and preparedness (like having spare tools, technicians on-site) is essential. After gaining experience, the operator's personnel are now much more familiar with the process, so the perceived complexity is reduced.
4. **Applicability and Limitations:** THO is most beneficial in scenarios where a separate reaming pass is costly or risky. This includes wells with troublesome formations (like presence of shale in salt) or wells where reaching TD in one go is highly desirable. In more benign formations, the incremental benefit may not justify the extra equipment. Also, not every well can accommodate two reamers. Small hole sizes or very long BHA tool strings might pose space constraints. In our operation, the hole was large (17½"), making it easier to include multiple large underreamers. In slim-hole drilling, tandem reamers would be much smaller and perhaps harder to design robustly. Another limitation to note is that directional drilling considerations must be accounted for: the presence of reamers can affect the BHA stiffness and steerability. In our vertical section, this was not a concern, but for highly deviated or complex trajectories, one must ensure the rotary steerable system (RSS) performance is not hindered. Fortunately, BHA could be enhanced and utilized with modern RSS tools can handle underreamers in the BHA after redesigning the hydraulic design.
5. **Economic Impact:** From a cost perspective, the time saved translates to rig cost savings (6-8 days per well compared to the planned duration). There is also value in **avoiding trouble costs** that might arise during separate hole opening (like stuck pipe or lost BHA). In fact, the THO tools **cost analysis was too positive** for subsequent wells in the project revealed that the operation saved approximately 8 rig days, equating to ~\$600k in spread cost savings, in addition to that this technique helps to avoid

cost of 3.9M of Utilizing normal underreams that has offered with \$1M per well (500K per run) and the service provider approach was to run it single not in tandem.

In summary, the discussion underscores that THO is not just a novelty but a **practical method with real well construction advantages**. It does require advanced planning and the right conditions to be effective. In cases where those conditions are met, as demonstrated in this study, the technique can deliver wells faster and with fewer problems. This aligns well with the operator's broader strategy of applying innovative technologies to improve efficiency (indeed, this project was part of a larger initiative to enhance drilling performance in the asset

Key Lessons Learned for Takeaways

Based on this trial, the following lessons are highlighted for applying THO in future projects:

Design Lessons Learned

- *Meticulous Planning Pays Off*: Invest time in pre-job modelling (torque/drag, hydraulics, dynamics) to tailor the dual-hole opening BHA for the specific well. Upfront engineering and tool testing were essential to our success.
- *MWD Optimization in Tandem Hole Opening BHA*: During the trial, MWD was dropped from the BHA due to pressure loss caused by fluid diversion across the hole openers. Since the section was vertical and guided by a bullnose, directional control was not critical making MWD removal a practical design simplification.
- *Ensure Tool Reliability*: Use proven underreamer designs and have contingencies, backup tools should be on hand.
- *Optimize Mud and Hydraulics*: The drilling fluid must handle increased cuttings load. High flow rates and appropriate rheology are needed for effective cleaning. Consider more inhibitive mud systems if formations are highly reactive, to preserve wellbore stability.
- *Hoel Opener Revolution limits*: as lessons learned author limit the tandem Krevs of this system to max of 550 Krev to avoid any associated risks of bearing failure or and complicate the operation.

Operational Lessons Learned

- *Real-Time Monitoring and Flexibility*: Continuous data monitoring allowed rapid adjustments to mitigate vibrations and optimize ROP. Ensure experienced personnel are monitoring downhole sensor data and empower them to adjust parameters on the fly.
- *Crew Training and Coordination*: A dual-reamer run introduces new procedures. Conduct thorough pre-job meetings and training so that drillers and rig crew know how to recognize proper tool function (pressure signatures, etc.) and handle any anomalies. Communication between office experts and rig site is vital during execution.
- *Contingency Planning*: Define clear contingency actions for scenarios like a reamer failing to open or unexpected torque spikes. Having a fallback plan (e.g., proceed with single reamer if needed) prevents indecision if something goes off script.
- *Capturing Performance Data*: After the run, perform detailed post-run analysis of vibration logs, tool wear, and wellbore condition. This helps validate design assumptions and improve the program for the next well (for instance, fine-tuning placement of reamers or stabilizers).
- *Post-Run Analysis*: Capture and analyse post-run performance data (vibration logs, tool wear, wellbore condition) to validate design assumptions and improve future runs.

Implementing these lessons will increase the likelihood of repeat success and help mainstream the tandem hole-opening technique in suitable candidates across the drilling portfolio.

Conclusion

The implementation of a tandem hole-opening approach in a problematic shale section resulted in a markedly improved drilling performance and validated the concept for broader use. Key conclusions from this case study are:

- **Feasibility:** A dual-reamer BHA can be successfully run and operated to enlarge a wellbore in a single drilling run. The necessary technical elements (Applicability and objective suitability, compatible BHA tools, etc.) are available and were proven to work reliably under field conditions. This trial was the first in the region and demonstrated that with proper engineering, the method is feasible even in onshore fields with conventional rigs.
- **Efficiency Gains:** The approach unlocked significant efficiency gains, cutting the section drilling time nearly in half. By avoiding an extra trip and integrating operations, the well was delivered ahead of schedule. This confirms the central hypothesis that THO can accelerate well delivery. In cost terms, the value of saved rig time and trouble avoidance is substantial, making the technique economically attractive.
- **Wellbore Quality and Stability:** Drilling and enlarging concurrently yielded an excellent wellbore. The continuous reaming stabilized the formation and prevented time-dependent degradation of the open hole. The operator was able to run casing smoothly to bottom and obtain a quality cement job, indicating that the wellbore condition was ideal. Thus, tandem reaming can be considered not only a time saver but also a risk mitigator for borehole stability in reactive or weak formations.
- **Tool Performance:** The dual hole openers and associated equipment's withstood the operation with minimal wear, suggesting that concerns about added tool strain are manageable. The distribution of cutting work across the two reamers and the damping of vibrations contributed to preserving tool integrity. No fishing or hole problems occurred, highlighting that a carefully planned dual-reamer run can be executed **safely without incident**.
- **Operational Considerations:** The successful outcome was aided by detailed planning, real-time monitoring, and adaptive adjustments. It is concluded that to replicate this success, future applications should ensure similar diligence including pre-job simulations, training of personnel, and having contingency plans. When these best practices are followed, the complexity of THO becomes comparable to standard operations, rather than a hindrance.
- **Broader Implications:** This case adds to a growing body of field evidence that smarter drilling practices like THO can have transformative effects on drilling performance. It showcases how innovative approaches pioneered in more technically challenging environments can be transferred to onshore operations to solve local challenges. For the Asset Owner and industry peers, the lesson is that embracing such technologies can lead to step-change improvements. In particular, for wells with hard or unstable sections where additional reaming runs are routine, **dual reamer technology offers a proven method to save time and reduce risk**.

In conclusion, the tandem hole-opening technique achieved its objectives in this trial, unlocking faster well delivery through a combination of sound engineering and operational execution. Going forward, the operator plans to utilize this approach on similar wells and will continue to refine the process (for instance, exploring its use in deviated holes or with different mud systems). The knowledge gained in terms of BHA design, real-time management, and hole cleaning is being integrated into the company's drilling programs. As the industry pushes for more efficient drilling to reduce costs and accelerate production, the positive

results from this case study serve as an encouraging example of innovation yielding tangible benefits. The authors advise operators with similar drilling challenges to use tandem hole-opening and work with service providers to customize the approach. What was once an experimental idea has now been field proven, and we anticipate wider adoption in the near future as part of the toolkit for efficient well construction.

Nomenclature

BHA	Bottomhole Assembly (the drilling tool string including bit, reamers, stabilizers, etc.)
ROP	Rate of Penetration (drilling speed, typically measured in meters per hour)
MWD	Measurement While Drilling (downhole tool for real-time directional and logging data)
RSS	Rotary Steerable System (directional drilling tool enabling steering without sliding)
ECD	Equivalent Circulating Density (effective fluid density in the well when pumping)
HEWD	Hole Enlargement While Drilling (general practice of enlarging borehole while drilling)
TD	Total Depth (final depth of the well or section)
HO	Hole Opening / Hole Opener (drilling a larger hole after a pilot hole)
THO	Tandem Hole Opening

References

1. Goodwin, A., Salminen, C., and Valverde, E. 2015. Dual Reaming Reduces Vibration in Deepwater Well. Paper SPE-174992-MS presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, 28–30 September.
2. Osama, M., Salem, G., Bin Sumaida, A.S. et al. 2024a. Case study of the revolutionary approach of the Middle East's first sustainable underground salt cavern oil storage: well design, drilling challenges, and mitigations. Paper SPE-222655-MS presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 4–7 November.
3. Osama, M., Salem, G., Bin Sumaida, A.S. et al. 2024b. Case study of the pioneering coring preservation and management of the first underground oil storage in the Middle East region. Paper SPE-222766-MS presented at ADIPEC, Abu Dhabi, UAE, 4–7 November.
4. Saheta, V. 2013. Dual-reamer technology eliminates cleanout trips. *E&P (Hart Energy)* **86** (12): 34–36 (December).
5. R. L. Loofbourow, Wm. G. Wood and R. T. Lassiter 1975. Construction Of Underground Storage Chambers. Presented at, Dallas, Texas, 15–16 May, SPE 5428.
6. Li, Research Institute of Petroleum Exp. & Dev.-Langfang, CNPC 1998. Research on utilization shallow structure to construct underground natural storage. Presented at Research Institute of Petroleum Exp, China, 2-6 Nov 1998, SPE 50935.
7. Design and Operation of Solution-mined Salt Caverns Used for Natural Gas Storage, *API Recommended Practice 1170* Second Edition, November 2022.
8. Functional Integrity of Natural Gas Storage in Depleted Hydrocarbon Reservoirs and Aquifer Reservoirs, *API Recommended Practice 1171 First Edition*, September 2015.
9. Storage Facilities Of Liquid Hydrocarbons In Jebel Dhanna Salt Dome Caverns, *Geostock-ADNOC*